

Section 5B. K-fold Cross-Validation

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K-fold Cross-validation (KCV)

Main Steps: Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, take the following steps:

1. Randomly partition the available data samples into K (almost) equal-sized parts: $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_K$. Denote the sizes by $N_1, N_2, \dots, N_K$
2. For $k = 1$ to K , do the following:
 - 2.1 Train the model using the data $\mathcal{D} \setminus \mathcal{D}_k$ (remove the k -th part):

$$\hat{\theta}_k = \arg \min_{\theta} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D} \setminus \mathcal{D}_k} (y_i - f(\mathbf{x}_i; \theta))^2$$

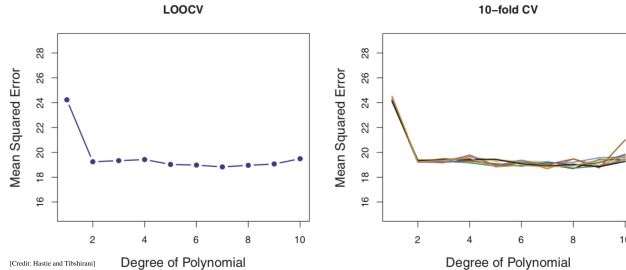
- 2.2 Predict the test MSE for the left-out part $\mathcal{D}_k$ using the formula

$$\text{MSE}_k = \frac{1}{N_k} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_k} \left(y_i - f(\mathbf{x}_i; \hat{\theta}_k) \right)^2$$

3. The K -fold cross-validation estimate is given by $\text{MSE}_{\text{KCV}} = \sum_{k=1}^K \frac{N_k}{N} \text{MSE}_k$

K-fold CV: Numerical Example

We use the N-fold (also called LOOCV) and the 10-fold CV approach to estimate the test MSE of polynomial models of various degrees in the Automobile dataset:



Advantages of K-fold CV over standard CV:

- ▶ Provides a less volatile estimate of the test MSE
- ▶ Uses more observations in training
- ▶ Provides a less overestimated test MSE

K-fold CV for Classification

Main Steps: Consider an available dataset $\mathcal{D}$ (with categorical outputs):

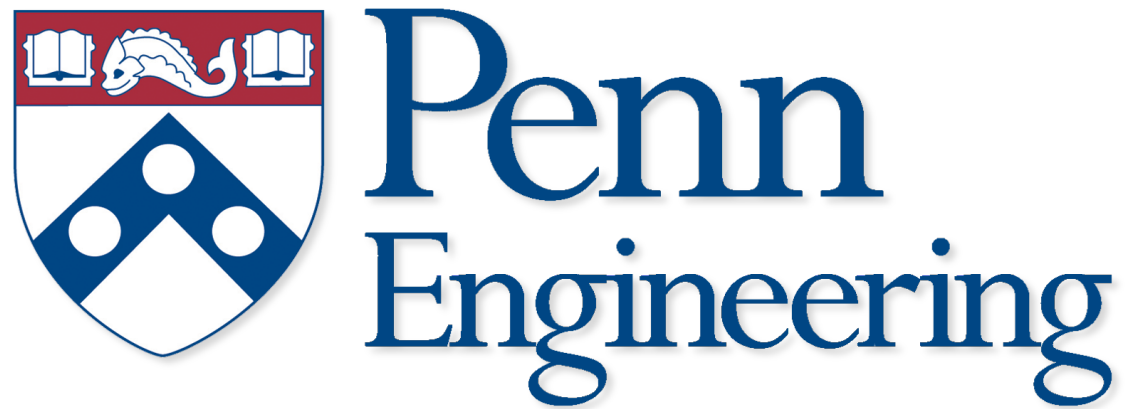
1. Randomly partition the available data samples into K (almost) equal-sized parts: $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_K$. Denote the sizes by $N_1, N_2, \dots, N_K$
2. For $k = 1$ to K , do the following:
 - 2.1 Train your classifier using the data $\mathcal{D} \setminus \mathcal{D}_k$ (remove the k -th part)

$$\hat{\theta}_k = \arg \max_{\theta} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D} \setminus \mathcal{D}_k} \delta(y_i - C(\mathbf{x}_i; \theta))$$

- 2.2 Estimate the misclassification error rate for the left-out part $\mathcal{D}_k$ using the formula

$$\text{Err}_k = \frac{1}{N_k} \left[N_k - \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_k} \delta(y_i - C(\mathbf{x}_i; \hat{\theta}_k)) \right]$$

3. The K -fold cross-validation misclassification error estimate is $\text{Err}_{\text{KF}} = \sum_{k=1}^K \frac{N_k}{N} \text{Err}_k$



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